

scripts/05_export_dashboard_data.py

71 lines

```
1 """
2 05_export_dashboard_data.py
3 =====
4 Bundles everything the dashboard needs (scenario logs, per-step MPC
5 reasoning, limits, controller config) into a single `dashboard/data.js` file
6 that defines `window.DASHBOARD_DATA`.
7
8 A plain <script src="data.js"> tag (not `fetch`) is used by the dashboard so
9 that it works when the HTML file is simply double-clicked and opened from
10 disk, with no local server and no network access required.
11 """
12
13 from __future__ import annotations
14
15 import json
16 import sys
17 from pathlib import Path
18
19 sys.path.insert(0, str(Path(__file__).resolve().parents[1] / "src"))
20
21 import pandas as pd
22
23 from dataclasses import asdict
24
25 from config import DATA_DIR, DASH_DIR, LIMITS, CONTROLLER, SCENARIOS, PLANT, TS_HOURS
26
27
28 def main():
29     fitted_model = json.loads((DATA_DIR / "fitted_model.json").read_text())
30
31     bundle = {
32         "limits": LIMITS.as_dict(),
33         "controller": {
34             "ts": CONTROLLER.ts,
35             "u_min": CONTROLLER.u_min,
36             "u_max": CONTROLLER.u_max,
37             "max_move": CONTROLLER.max_move,
38             "horizon": CONTROLLER.horizon,
39             "candidate_step": CONTROLLER.candidate_step,
40             "move_penalty": CONTROLLER.move_penalty,
41             "margin_whp": CONTROLLER.margin_whp,
42             "margin_flp": CONTROLLER.margin_flp,
43             "margin_bhp": CONTROLLER.margin_bhp,
44             "bias_filter_alpha": CONTROLLER.bias_filter_alpha,
45         },
46         "plant": asdict(PLANT),
47         "ts_hours": TS_HOURS,
48         "model": fitted_model,
49         "scenarios": {},
50     }
51
52     for key in ("A", "B", "C"):
53         df = pd.read_csv(DATA_DIR / f"scenario_{key}_log.csv")
54         decisions = json.loads((DATA_DIR / f"scenario_{key}_decisions.json").read_text())
55
56         bundle["scenarios"][key] = {
57             "name": SCENARIOS[key]["name"],
58             "description": SCENARIOS[key]["description"],
59             "u_initial": SCENARIOS[key]["u_initial"],
60             "log": df.round(2).to_dict(orient="records"),
61             "decisions": decisions,
62         }
63
64     out = DASH_DIR / "data.js"
65     out.write_text(f"window.DASHBOARD_DATA = " + json.dumps(bundle) + ";\n")
66     print(f"wrote {out} ({out.stat().st_size/1024:.0f} KB)")
67
```

```
68
69 if __name__ == "__main__":
70     main()
71
```